

Sensor-Based Simulation

Project Title: Automatic Light Control Using LDR Sensor

Objective:

To automatically control an LED based on surrounding light intensity.

Components Used:

- Arduino Uno
- LDR (Photoresistor)
- LED
- 10k Ω Resistor
- 220 Ω Resistor
- Breadboard
- Jumper Wires

Connections

LDR Circuit

- One leg of LDR $\rightarrow$ 5V
- Other leg of LDR $\rightarrow$ A0 and 10k Ω resistor
- Other end of 10k Ω resistor $\rightarrow$ GND

LED Circuit

- LED Positive (Long Leg) $\rightarrow$ Pin 7
- LED Negative $\rightarrow$ 220k Ω Resistor $\rightarrow$ GND

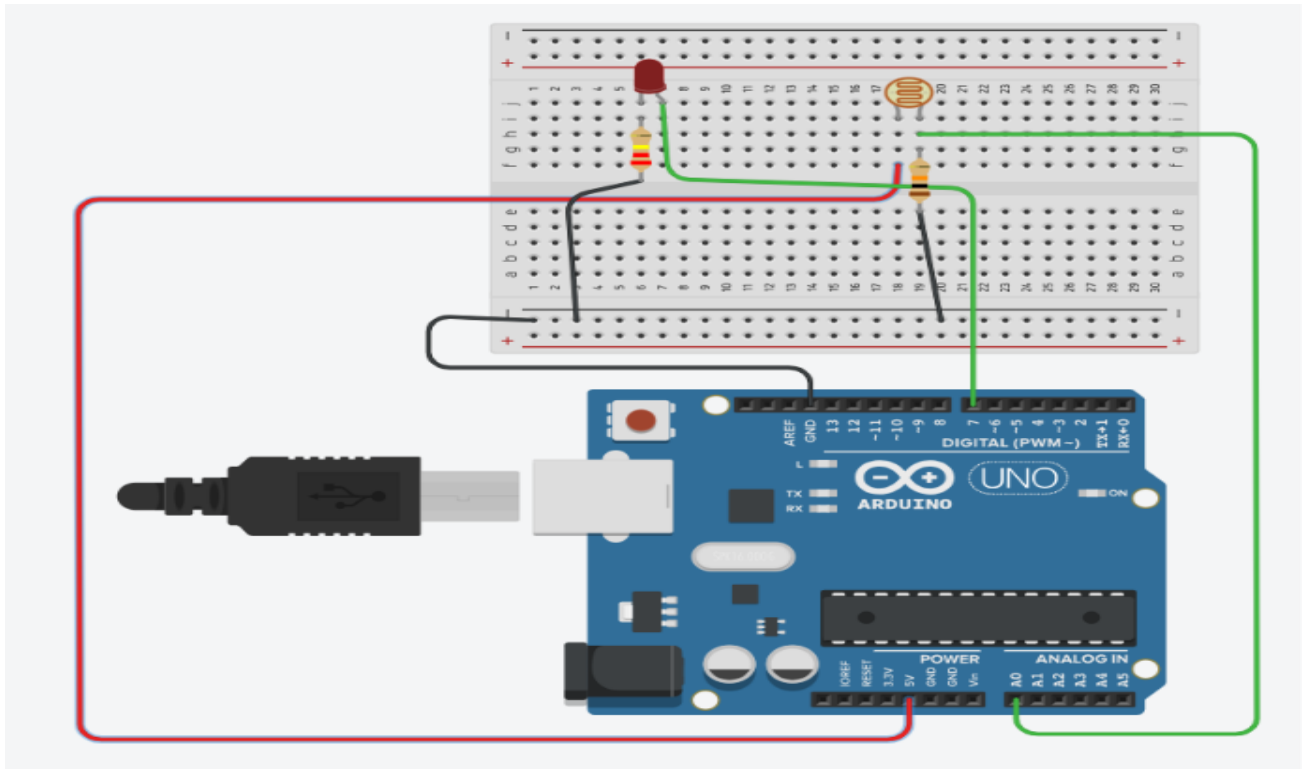


Figure 1: Circuit Diagram of Automatic Light Control Using LDR Sensor

ARDUINO CODE:

```
1 // C++ code
2 //
3
4 int ldrPin = A0;          // LDR connected to A0
5 int ledPin = 7;          // LED connected to pin 7
6 int threshold = 500;     // Adjust if needed
7
8 void setup() {
9     pinMode(ledPin, OUTPUT);
10    Serial.begin(9600);
11 }
12
13 void loop() {
14     int ldrValue = analogRead(ldrPin);
15
16     Serial.print("LDR Value: ");
17     Serial.println(ldrValue);
18
19     if (ldrValue < 500) {
20         digitalWrite(ledPin, HIGH); // LED ON
21     } else {
22         digitalWrite(ledPin, LOW);  // LED OFF
23     }
24
25     delay(100);
26 }
```

Figure 2: Arduino Program for LDR-Based Automatic Light Control

Code Explanation

- The LDR sensor is connected to analog pin A0 and the LED is connected to digital pin 7.
- The Arduino reads the light intensity from the LDR continuously.
- The LDR value is displayed on the Serial Monitor for observation.
- A threshold value of 500 is used to determine light and dark conditions.
- If the LDR value is less than 500 (dark environment), the LED turns ON.
- If the LDR value is greater than or equal to 500 (bright environment), the LED turns OFF.
- The program repeats this process continuously with a small delay of 100 milliseconds.
- This project demonstrates an automatic light control system using an LDR sensor and Arduino.

How It Works

1. LDR senses light intensity.
2. Arduino reads the sensor value through pin A0.
3. When light decreases (dark room), the sensor value decreases.
4. Arduino turns ON the LED.
5. When light increases, Arduino turns OFF the LED.

in Tinkercad

1. Open Tinkercad → Circuits.
2. Create a New Circuit.
3. Add Arduino Uno, LDR, LED, resistors, and breadboard.
4. Connect the components.
5. Paste the Arduino code.
6. Click **Start Simulation**.
7. Change the LDR light level slider.
8. Observe the LED turning ON/OFF.

Simulation Output:

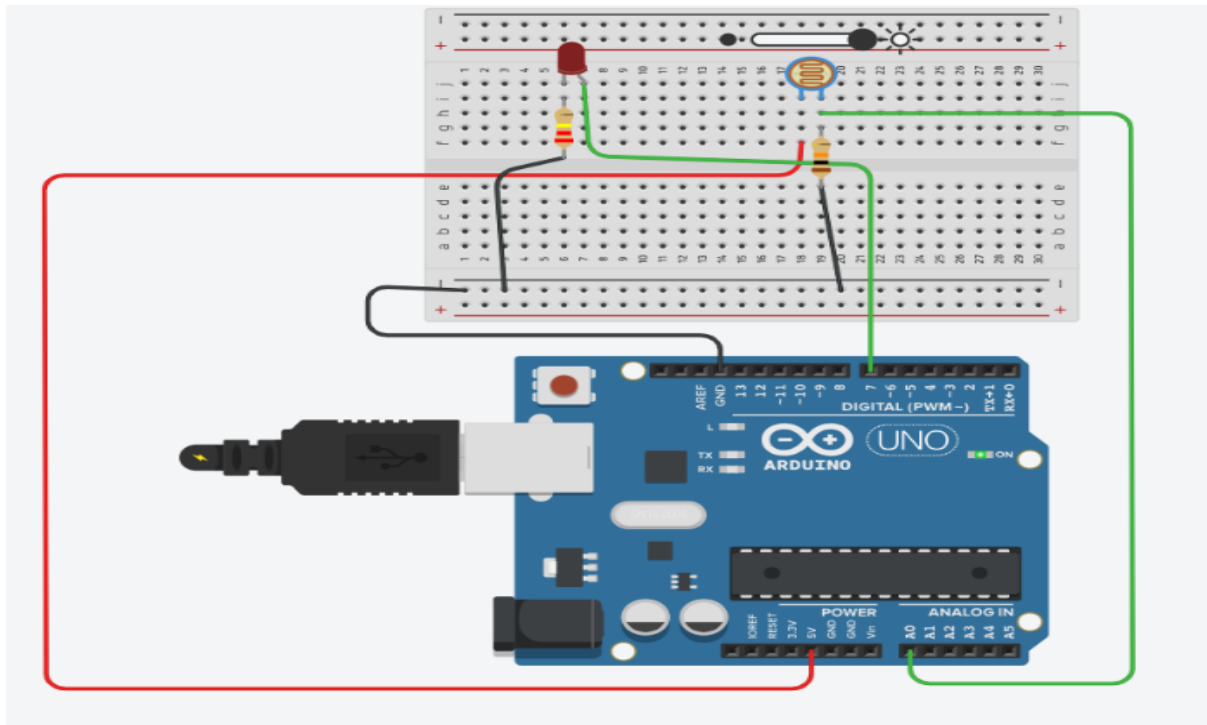


Figure 3: Output Under Bright Light Condition (LED OFF)

Output Observation 1: Bright Light Condition

- The LDR receives sufficient light.
- Arduino reads a higher sensor value.
- The LED remains OFF.
- This indicates that the system detects adequate ambient light.

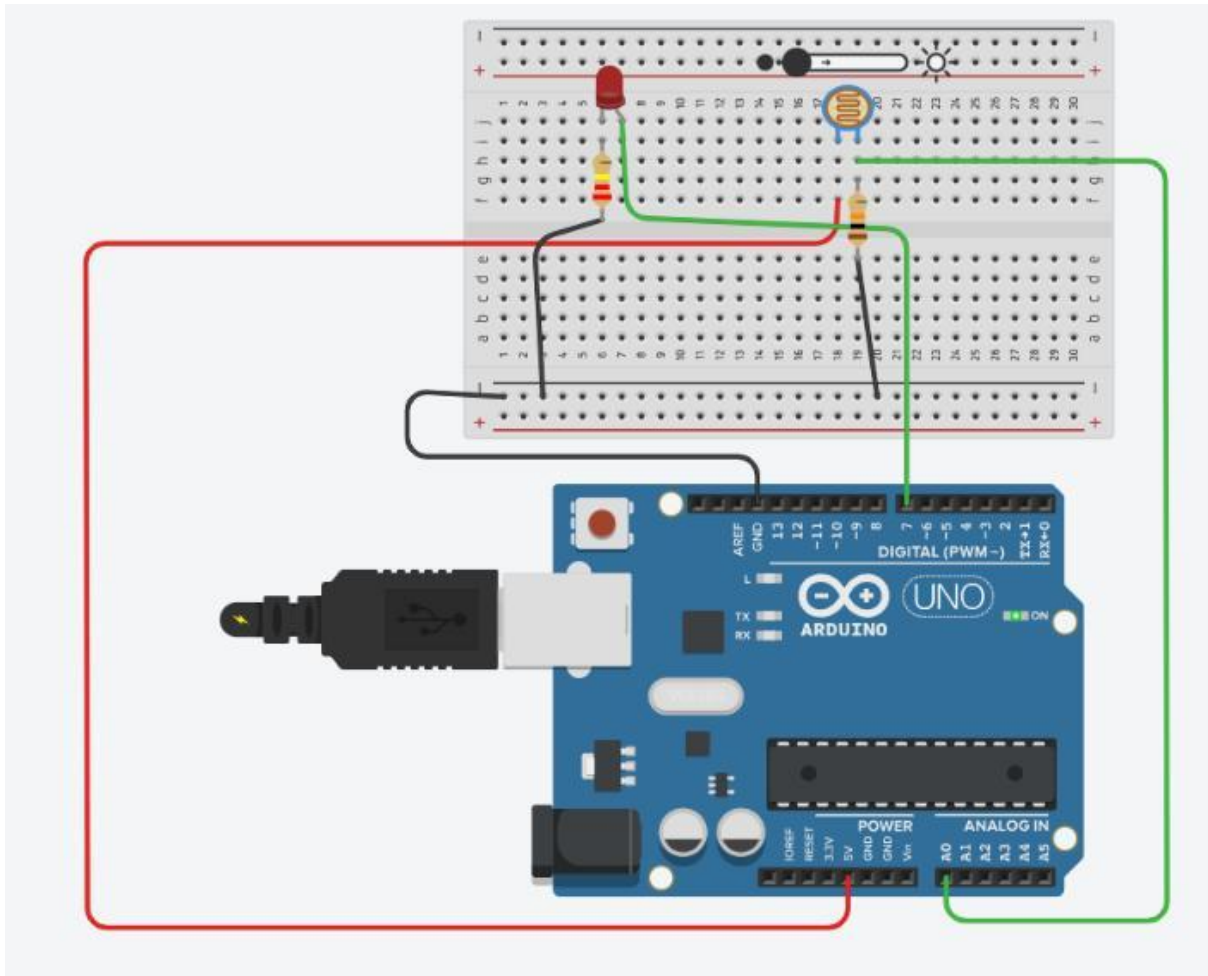


Figure 4: Output Under Dark Light Condition (LED ON)

Output Observation 2: Dark Light Condition

- The LDR receives less light.
- Arduino reads a lower sensor value.
- The LED turns ON automatically.
- This demonstrates automatic lighting control in low-light environments.

Output Analysis

- The system successfully responds to changes in light intensity.
- The LED state changes according to the LDR sensor readings.
- The project can be used for energy-efficient lighting applications.
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Result

The Automatic Light Control System using an LDR Sensor was successfully simulated in Tinkercad. The LED turned ON in darkness and OFF in bright light, achieving the desired objective.